

DENZEL JOSHUA MOFFAT

Data Science | Machine Learning | Software Development

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PROFESSIONAL SUMMARY

Computer Science student at the University of London with practical experience building end-to-end data and software projects in Python, SQL and Node.js, spanning supervised machine learning, natural language processing, relational database design and full-stack web applications. Experienced in tutoring and first-line IT support. Seeking internship, attachment or graduate roles in data science, data analytics or software development.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript (ES6+), C++, HTML5, CSS3

Data & Machine Learning: pandas, NumPy, scikit-learn, Matplotlib, Seaborn, SciPy; regression, classification, cross-validation, feature engineering, model evaluation

Natural Language Processing: NLTK, TF-IDF, GloVe word embeddings, tokenisation, lemmatisation, stemming

Databases: MySQL, SQLite; ER modelling, normalisation (1NF to BCNF), joins, aggregation, ETL pipelines

Web & Tools: Node.js, Express, EJS, Git/GitHub, Jupyter, VS Code, Linux

EDUCATION

BSc (Hons) Computer Science — University of London (Goldsmiths) | 2024 – 2026 (expected)

Selected module marks: Software Design and Development 75 • Computational Mathematics 73 • How Computers Work 73 • Computer Security 72

A-Levels, Golden Springs, Harare (2023): 13 points | O-Levels, Westminster International School, Harare (2021): 9 A's

PROJECTS

Fake News Detection: Statistical vs Embedding Models | *Python, scikit-learn, NLTK, GloVe*

- Compared two text-classification pipelines on a 44,898-article corpus, holding the classifier and stratified 80/20 split constant so performance differences isolated the feature representation.
- Achieved 99% accuracy and 0.99 macro-F1 with TF-IDF and Logistic Regression, against 96% for averaged 100-dimensional GloVe embeddings, benchmarked on majority-class and Naive Bayes baselines.
- Identified that the headline result partly reflected stylistic artefacts in the source data rather than generalisable signal, and documented the limitation.

Football Player Market Value Prediction | *Python, pandas, scikit-learn*

- Decomposed a 10-million-row dataset into three normalised tables, de-duplicated to 197,000 records, and applied a log transformation justified by measured skew and kurtosis.
- Trained a log-linear regression reaching R^2 0.974 under 5-fold cross-validation, with mean absolute error of roughly €690,000 on the real currency scale.
- Tested and rejected a degree-2 polynomial model after it collapsed on inverse-log back-transformation, selecting the simpler model on the evidence.

Scouting Database | *MySQL, Node.js, Express, EJS*

- Designed a seven-entity relational schema normalised to BCNF, resolving a many-to-many relationship into an associative entity to preserve report attributes.
- Wrote a custom Node.js ETL script using streamed CSV parsing and async sequencing after native MySQL import failed, handling type coercion and null standardisation.
- Authored analytical SQL queries using multi-table joins and aggregation to quantify wage and valuation disparities across European leagues.

Event Booking and Ticketing Platform | *Node.js, Express, SQLite, EJS*

- Built a three-tier web application with routing separated by user role for event organisers and attendees.
- Implemented server-side ticket availability validation and booking persistence with foreign-key integrity.
- Added a revenue reporting page aggregating sales per event using LEFT JOIN, SUM and COALESCE.

Additional: C++ time-series analysis toolkit with custom CSV parsing and rolling-average trend calculation; JUCE multi-deck DJ application with a real-time audio mixing pipeline.

EXPERIENCE

Private Tutor — Computer Science, Mathematics, Physics and Chemistry | *[2024 - Ongoing]*

- Taught O-Level and A-Level students across four subjects, adapting explanations and worked examples to each student's level and pace.
- Broke down abstract programming and mathematical concepts into step-by-step reasoning to build independent problem-solving.

IT Support | *[Church, Digital solutions]* | *[2025-2026]*

- Diagnosed and resolved everyday hardware and software faults for end users.
- Guided users unfamiliar with computing through device setup and routine tasks in plain language.

LEADERSHIP AND ACTIVITIES

Lesson lead and teacher, church study programme, planned and delivered lessons to regular groups. School Prefect and Sports Captain: held appointed responsibility for peers and team organisation.